

Scalar tensor products do not add PI exponents

Disproof of Conjecture 00000002751

zhangni-sys — October 8, 2026

Counterexample. Take the unital characteristic-zero PI algebras $A = B = \mathbb{Q}$ over $\mathbb{Q}$. Their actual algebra tensor product is $T = \mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{Q}$. Multiplication defines an algebra isomorphism

$$\phi : T \longrightarrow \mathbb{Q}, \quad \phi(a \otimes b) = ab, \quad \phi^{-1}(c) = 1 \otimes c.$$

The inverse identity follows from the balancing relation $a \otimes b = 1 \otimes ab$. In particular T is a genuine nonzero commutative algebra, and each of A, B, T satisfies the nonzero commutator polynomial $xy - yx$.

Codimensions from evaluations. Let D be any nonzero unital commutative $\mathbb{Q}$ -algebra, in particular any of A, B, T . For $m \geq 1$ let P_m be the vector space of formal noncommutative multilinear polynomials, with basis $w_\sigma = x_{\sigma(1)} \cdots x_{\sigma(m)}$ for $\sigma \in S_m$. Its actual evaluation map on all tuples in D is

$$E_{D,m} : P_m \rightarrow \{D^m \rightarrow D\}, \quad E_{D,m} \left(\sum_{\sigma} a_{\sigma} w_{\sigma} \right) (t) = \sum_{\sigma} a_{\sigma} \prod_{j=1}^m t_{\sigma(j)}.$$

The kernel is exactly $P_m \cap \text{Id}(D)$. Commutativity shows that

$$E_{D,m}(p)(t) = \left(\sum_{\sigma} a_{\sigma} \right) g_m(t), \quad g_m(t) = \prod_{j=1}^m t_j.$$

The function g_m is nonzero since its value at $(1, \dots, 1)$ is $1_D \neq 0$. It belongs to the image because it evaluates any basis word. Hence the image is precisely its one-dimensional $\mathbb{Q}$ span. The first isomorphism theorem gives the genuine codimensions

$$c_m(D) = \dim_{\mathbb{Q}}(P_m / (P_m \cap \text{Id}(D))) = \dim_{\mathbb{Q}} \text{im } E_{D,m} = 1.$$

Consequently the defining n th-root limits exist and

$$\exp(A) = \exp(B) = \exp(T) = \lim_{m \rightarrow \infty} \sqrt[m]{1} = 1.$$

Thus the conjectured exact sum would assert $1 = 1 + 1 = 2$, a contradiction. The extra mixed-characteristic correction does not repair a false assertion already in characteristic zero. We only need this false clause to disprove the conjecture.

Formal certificate. Lean constructs the actual algebra tensor product and its standard multiplication algebra equivalence. The multilinear evaluation map, its quotient-by-kernel linear equivalence, one-dimensional image, codimensions and root limit are established for a general nonzero commutative $\mathbb{Q}$ -algebra and instantiated separately on the scalar factors and the tensor algebra. No exponent or codimension sequence is stipulated. Final theorem audits use only standard Lean and Mathlib axioms.